Abstract

The Movie Explorer is a web-based application developed to provide a structured and accessible platform for browsing, searching, and exploring a collection of movies. The primary objective of the project is to demonstrate the practical application of fundamental front-end web development concepts while providing users with efficient mechanisms for locating and examining movie information. The system is designed to organize movie data and support interactive exploration through a simple and responsive user interface.

The application is designed to make movie exploration convenient by presenting movie information in an organized and user-friendly manner. Users can browse the available collection and interact with the displayed movie information through a simple interface. The system focuses on reducing the effort required to locate specific movies and provides an effective way to examine the available collection.

The project is implemented exclusively using HTML, CSS, and JavaScript. HTML is used to define the semantic structure and organization of the application, while CSS is employed for presentation, layout, and responsive design. CSS Grid and Flexbox are used to create an adaptive interface that accommodates different screen sizes. JavaScript provides the application's dynamic functionality by managing movie data through arrays and objects, manipulating the Document Object Model (DOM), and implementing event-driven programming to respond to user interactions.

The application provides major features including movie search by title, genre-based filtering, sorting by rating and release year, and access to detailed movie information through interactive components. Users can search for a particular movie, filter movies according to genre, and organize the collection based on rating or release year. User actions dynamically update the displayed movie collection without requiring changes to the underlying page structure.

The resulting application demonstrates the integration of structural, presentation, and behavioral aspects of client-side web development within a single system. It provides practical experience in data handling, DOM manipulation, responsive interface development, and interactive programming. The project also establishes a foundation for future extensions, such as incorporating additional movie data and developing more advanced functionality while retaining the core front-end technologies.

24981A428P Tagaram Sonalika (TL)
24981A426U Chenna Varun
24981A425I Mamidi Rishith Kumar
24981A424Y Gutti Indu Valani
24981A426X Chodisetti Tejeswararao

Guide Signature **HOD Signature**